Detailed Usage Documentation for auto-feature-regression

💡 **Summary**: An automated feature engineering and linear regression modeling skill for regression prediction tasks.

Given the business background and analysis objectives, with an input of a raw dataset, it will automatically generate a report, visual charts and a model covering preprocessing, feature engineering, modeling and evaluation.

auto-feature-regression (5).zip
34.81KB

I. Usage Scenarios

Typical Scenarios	Explanation
Business Metric Forecasting	Predict continuous numerical targets such as quarter revenue, order volume, GMV, cost, and conversion rate, and identify key driving factors.
Feature Engineering Exploration	Want to quickly test interaction terms, polynomials, and time-series derived features, and automatically filter out meaningless, collinear, and non-informative features
time series regression	For business data with a date column, the training/test sets must be strictly divided by time to prevent future information leakage
Fast Interpretable Baseline	A linear model with interpretable coefficients, p-values and confidence intervals is required as the baseline, rather than a black-box model.
Analytical Report Generation	Need a report that includes descriptive statistics, correlation heatmaps, descriptions of redundant features, model formulas and metrics.

💡 **Limitations:**

It can only fit a linear regression model, and all features finally included in the model are required to be numerical features.

It is not applicable to classification tasks (where the target is a category), deep models for pure time-series prediction, and unstructured data (images/text/audio).

II. Input / Workflow / Output

2.1 Input Required from Users

Required parameter

item	Parameter	Explanation
Dataset File	`--data`	CSV / Excel, containing 1 numerical target column + several feature columns
Target Field Name	`--target`	Name of the numerical column to be predicted

Key Optional Parameters

Parameter	When to Use	Example
`--time-col`	When the dataset is time-series data, this parameter is mandatory; otherwise, it will degenerate into random partitioning, leading to distorted evaluation results.	`--time-col date`
`--exclude`	Exclude non-feature columns such as ID/name, or target-derived columns that are prone to leakage	`--exclude id,shop_name`
`--context`	A one-sentence business context to be included in the report and make the diagram more relevant	`--context "Quarterly Advertising Revenue Data"`
`--goal`	State the modeling objective in one sentence to keep the diagram focused on the objective	`--goal "Forecast quarterly ad revenue and identify driving factors"`
`--optimize-metric`		

	Specify the optimization metric to guide the feature selection direction	WMAPE (default)/MAE/RMSE/R2/Adj_R2
`--feature-mode`	Feature Construction Pattern	auto (default)/custom/both
`--custom-features`	Custom Business Feature Formula	`"revenue_intensity = revenue_L30 / 30"`
`--interaction-pairs`	When you know which feature combinations carry business meaning, specify them precisely	`"price*volume;ad_cost*ctr"`
`--impute` / `--impute-overrides`	Missing Value Imputation Strategy (Global / Column-wise)	`--impute-overrides "recharge:zero"`
`--log-target`	Log1p transformation switch for severely right-skewed targets	auto (default)/on/off
`--outdir`	output directory	`--outdir ./output`

> 📝 **Dependencies need to be installed for the first use:**
> `pip install --user --break-system-packages pandas numpy scikit-learn matplotlib seaborn statsmodels tabulate`

Example of invoking a command — time series (must pass `--time-col`)

时间序列建模

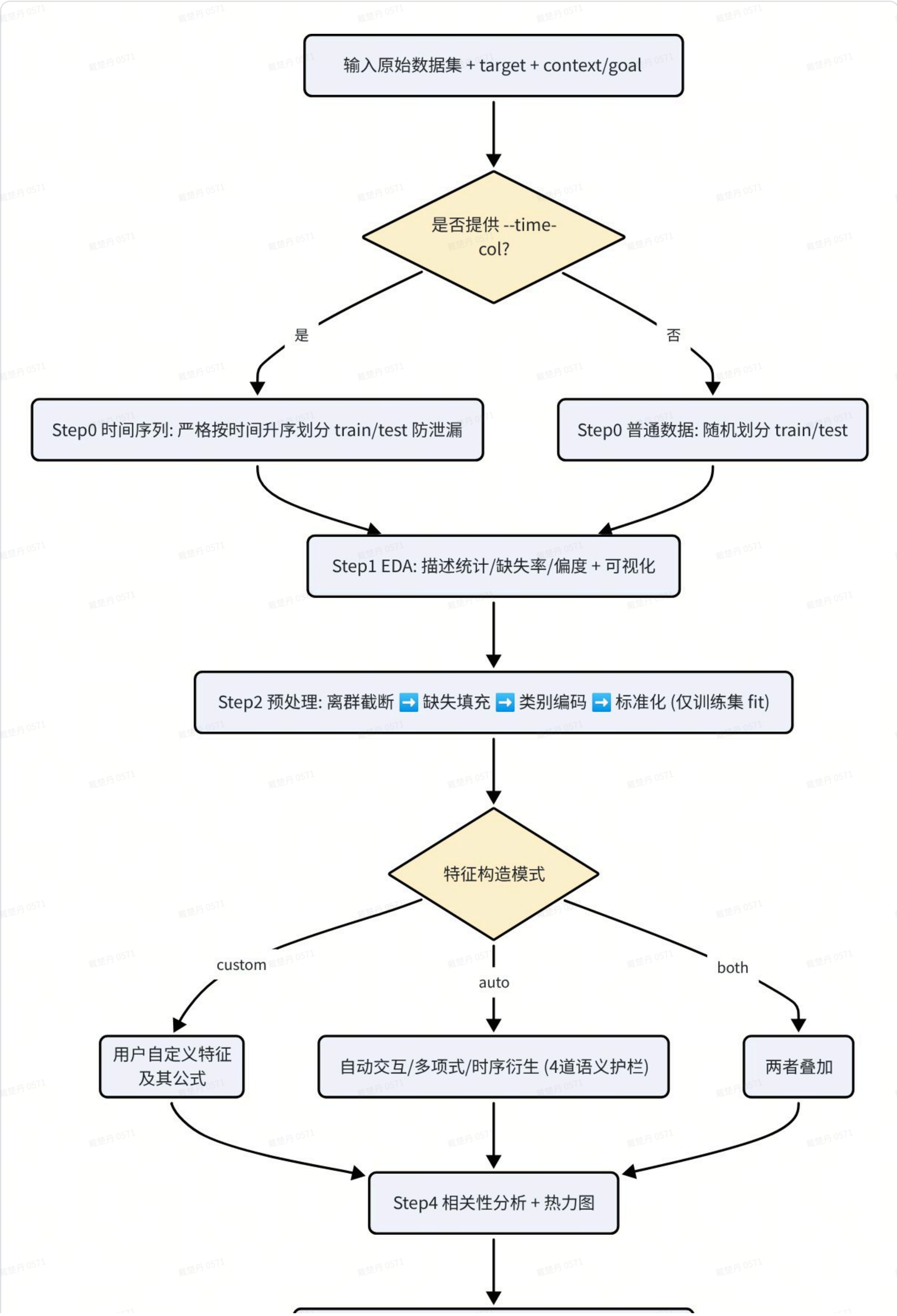
```
python3 scripts/auto_feature_regression.py \
    --data ~/files/sales.csv --target quarter_revenue \
    --time-col period_start_date --exclude account_id,account_name \
    --context "广告季度收入数据" --goal "预测季度广告收入并找驱动因子" \
    --optimize-metric WMAPE --outdir ./output
```

Example of invoke command —— Common section data (without passing `--time-col` → random partitioning)

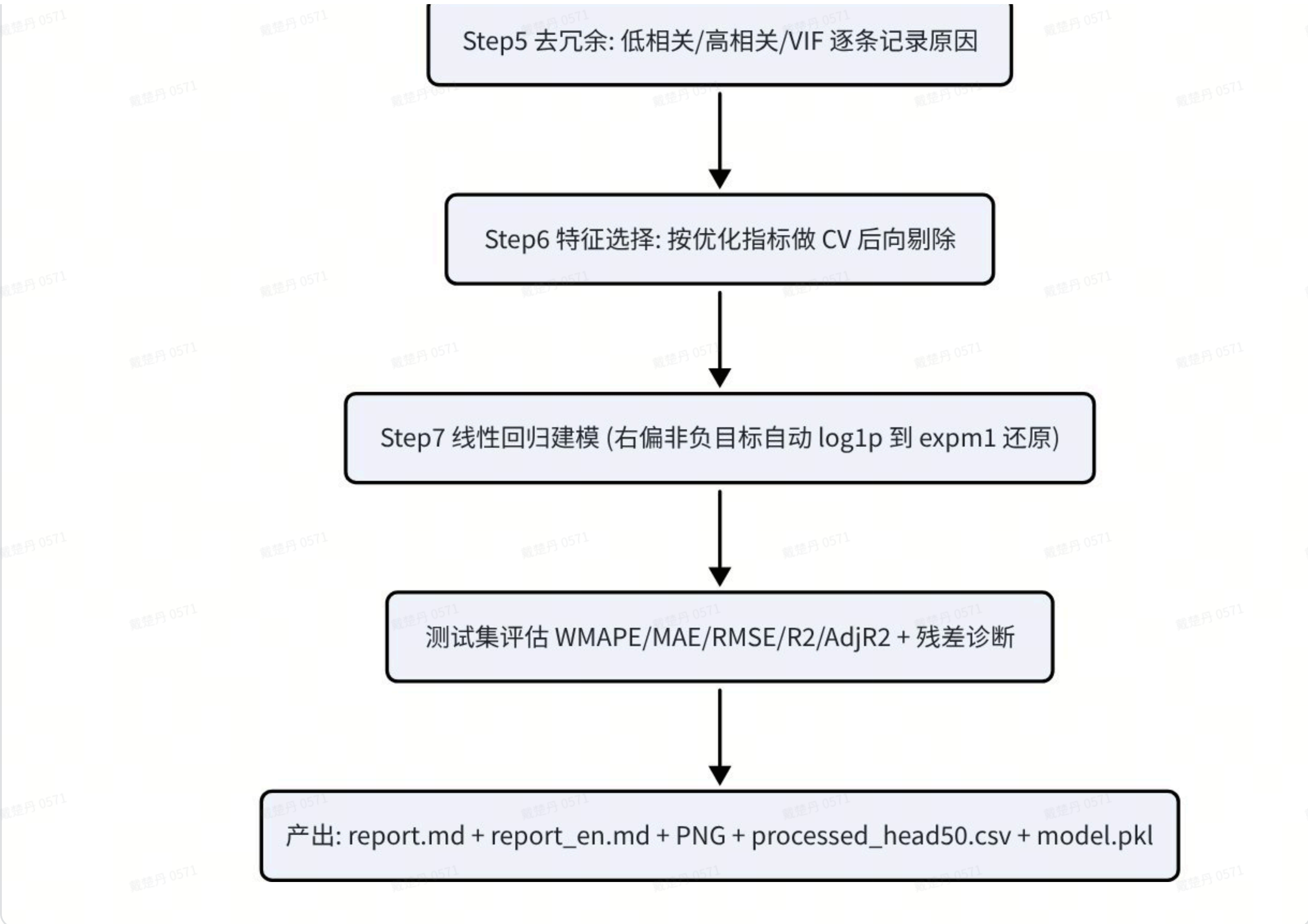
普通数据建模

```
python3 scripts/auto_feature_regression.py --data data.csv --target y --outdir ./output
```

2.2 Workflow



输入原始数据集 + target + context/goal
是否提供 --time-col?
是
否
Step0 时间序列: 严格按时间升序划分 train/test 防泄漏
Step0 普通数据: 随机划分 train/test
Step1 EDA: 描述统计/缺失率/偏度 + 可视化
Step2 预处理: 离群截断 ➡️ 缺失填充 ➡️ 类别编码 ➡️ 标准化 (仅训练集 fit)
特征构造模式
custom
auto
both
用户自定义特征及其公式
自动交互/多项式/时序衍生 (4道语义护栏)
两者叠加
Step4 相关性分析 + 热力图



7-Step Detailed Guide

Step	Content
Step 0 Data Partitioning	Provide `--time-col` → strictly split in ascending chronological order (without shuffling) to prevent future information leakage; otherwise, perform random splitting
Step 1 EDA	Mean / Variance / Extremum / Skewness / Missing Rate + Target Distribution Plot or Time Series Plot
Step 2 Preprocessing	Outlier quantile clipping (1% / 99%) → **Do not delete columns due to high missing rate**, suggest autofill strategies based on the distribution of each column (median / mean / mode / fill with 0) or directly force all missing values to be filled with 0 → One-Hot encoding → Standardization. All fit parameters are derived solely from the training set
Step 3 Feature Construction	`auto` automatically generate interactions, polynomials, and time-series derivatives (with four guardrails applied to interaction terms); `custom` business formulas provided by the user; `both` superposition. All constructed features are registered, and the reason for each skipped combination is recorded one by one

Step 4 Correlation Analysis	Correlation Heatmap + Feature-Target Correlation Bar Chart
Step 5 Redundancy Removal	Low correlation with target / high correlation between features / VIF multicollinearity, record the deletion reason for each item
Step 6 Feature Selection	Based on `--optimize-metric` , backward elimination is performed after cross-validation on the training set (both temporal TimeSeriesSplit and ordinary KFold use 4 folds).
Step 7 Modeling Evaluation	Linear regression + test set metric + residual diagnosis; automatic log1p transformation for severely right-skewed non-negative targets, expm1 restoration after prediction, with metrics calculated on the original explicitness

📌 **Data leakage prevention is the top principle**: All preprocessing parameters (fill values, truncation quantiles, encoding, standardization) are fitted only on the training set before being applied to the test set; time series are strictly split by time without any shuffling.

2.3 Final Output (Located in --outdir)

File	Content
`report.md`	Analyze background and objectives, preprocessing summary, descriptive statistics, correlation, redundancy removal rationale, constructed feature list, final features, model formula + coefficient table (including p-values / confidence intervals), evaluation metrics, and data partitioning description;**Each figure is accompanied by a data-driven "graphical interpretation"**
`report_en.md`	The English version of the aforementioned report has an identical structure (sharing the same charts, formulas and numerical values, with only the narrative and table headers translated), and the two versions use the same set of figures.
`01_missing_rate.png` and other PNG files	Missing rate, target distribution / time series, correlation heatmap, feature-target correlation bar chart, residual diagnostic plot; titles / coordinate meanings / units are all labeled, large values are automatically formatted in K/M/B, and automatically fallback to English if no Chinese font is available
`processed_head50.csv`	The first 50 rows of the preprocessed data for easy verification

`model.pkl`	The fitted model, scaler, feature list and preprocessing parameters are provided for subsequent reuse or prediction.

2.4 Example

III. Iteration Process Record

0723 Iteration

3.1 Explanation of Charts

Combine `--context` / `--goal` with the real statistics of this graph (mean / skewness / strongest correlation / signs of heteroscedasticity) to explain what this graph means for the analysis objective.

3.2 Apply log transformation to the target (--log-target)

> ❓ **Judgment Logic (auto mode)**
>
> - `--log-target on` → Force log1p
> - `--log-target off` → Force no transformation
> - `--log-target auto` (default) → When **the target skewness > 1.0 and the target minimum value ⩾ 0**, perform log1p transformation, apply expm1 for restoration after prediction, and compute all metrics on the original explicitness; otherwise, no transformation is applied.

Why is it designed this way? For targets such as revenue, GMV, and costs that are heavily right-skewed and non-negative, taking the logarithm makes their distribution closer to normal and better fits the homoscedasticity assumption of linear regression, which can significantly improve the fitting effect and residuals. This function will be automatically disabled when there are negative values (as log1p is undefined), ensuring robustness.

3.3 Construction of Interaction Term Features

In the original version, blindly multiplying features in pairs would generate a large number of features that have no business meaning and exacerbate collinearity. Now, the `auto` mode sequentially gates each candidate interaction term `a×b`, and the reasons for the skipped combinations are recorded in the report one by one:

		Rules and Thresholds
①	homology filtering	`INTER_SKIP_SAME_FAMILY= True`: identifies "different periods/shards of the same metric" based on the root of column names (e. g. both revenue_q1 and revenue_q2 have the root revenue), and by default such columns will not be multiplied — multiplication of the same type usually has no business meaning. `--allow-same-family` can be used to lift this restriction.
②	High Collinearity Filtering	`INTER_SKIP_HIGH_CORR= 0.8`: Do not multiply features when their correlation coefficient is greater than 0.8, as multiplication will only further exacerbate multicollinearity
③	Gain Verification	`INTER_MIN_GAIN= 0.02`: The |corr| between the interaction term and the target must be at least 0.02 higher than the |corr| of each of the two parent features; otherwise, it will be discarded as having no information gain
④	quantity cap	`INTER_MAX_KEEP= 8`: Only keep the top 8 entries sorted by gain to prevent dimensional explosion

> 💡 If the user knows exactly which pairs of variables carry business meaning when multiplied, use `--interaction-pairs "price*volume; ad_cost*ctr"` for precise specification → **skip all automatic combinations and guardrails**, and only generate the pairs listed by the user.

3.4 Missing Value Processing

- **No longer delete columns due to high missing rate**: For a missing rate exceeding `HIGH_MISS_HINT (= 0.5)`, only a prompt will be given, and the column will be retained for manual judgment of information value.
- `--impute auto` automatically suggests an imputation strategy (median / mean / mode / zero-filling) based on the distribution characteristics of each column, including skewness, non-negativity, zero proportion and cardinality. You can use `--impute` to specify a global strategy, or use `--impute-overrides` to override the strategy on a per-column basis.

3.5 Redundancy Removal and Feature Selection Rules

- **Low Correlation Elimination**: Eliminate candidates whose |r| with target is less than `CORR_TARGET_MIN (= 0.03)`.

- **High Correlation Elimination**: For two features where |r| > `CORR_HIGH (= 0.9)`, the one more correlated with the target is retained.
- **VIF Elimination**: Iteratively remove features with VIF greater than `VIF_THRESH (= 10.0)`.
- **CV backward elimination**: based on `--optimize-metric`, try removing one feature at a time; if the cross-validation metric can be improved, remove the feature, and repeat until no further improvement can be achieved.

IV. Future Optimization Directions

> ⭐ To adapt to more scenarios, this skill can be further enhanced in the following directions in the future.

1. **Model Family Expansion**: Add Ridge / Lasso / ElasticNet (with automatic regularization and built-in feature selection) as well as interpretable tree models (such as LightGBM with SHAP) as optional baselines to compare the linear and nonlinear fitting capabilities.
2. **Support for Classification/Counting Tasks**: Extended to logistic regression, Poisson/negative binomial regression, covering binary classification, ratio, and count-type targets, so that skills are no longer limited to continuous regression.
3. **Enhanced feature engineering**: added binning/target encoding, periodic time features (one-hot encoding for weekday/month/holiday), multi-order (ternary) interactions, and automatic semantic grouping based on business dictionaries.
4. **Automatic Tuning of Hyperparameters and Thresholds**: Treat `Config` the truncation quantile, VIF/correlation threshold, and interaction gain threshold as objects for grid/Bayesian search, and set them adaptively based on data instead of using fixed values.
5. **Data Quality and Drift Detection**: Added training/test distribution drift detection (PSI/KS), outlier labeling, and automatic target leakage detection (automatic alert for features with abnormally high correlation with the target).
6. **Cross-Validation and Robustness Enhancement**: Supports grouped cross-validation (by store/user), nested cross-validation, and time-series rolling prediction evaluation; outputs prediction intervals instead of merely point estimates.
7. **Interactive and Reusable Artifacts**: Output interactive HTML reports, one-click prediction scripts (which load `model. pkl` to score new data), and RESTful prediction interface templates.

V. Statistical Knowledge Involved

💡 This section covers the statistical knowledge involved in the entire skill workflow: conceptual definitions, **calculation formulas**, **business selection recommendations**, and **chart interpretation methods**, to help readers without a statistics background understand every metric and judgment in the report.

Assumptions of Linear Regression: LINE

🥛 L stands for Linear, which means the relationship between the dependent variable and the independent variables should satisfy the linearity assumption; I stands for Independent, which means the independent variables are mutually independent, the residuals are mutually independent, and the residuals are independent of the independent variables; N stands for Normality, which means the distribution of the residuals should follow a normal distribution; E stands for Equal Variance, which means the variance of the residuals is a constant.

5.1 Skewness and log Transformation

- **Skewness**: A measure of the asymmetry of a distribution that is approximately normal**.** A skewness value > 0 indicates right skew (long tail on the right, with a small number of large values, e. g., income / GMV), < 0 indicates left skew, and a value close to 0 indicates approximate symmetry.
- **Formula**: $\mathrm{Skew} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{\sigma}\right)^3$, i. e., the third-order moment after standardization.
- **Rationale**: Linear regression assumes that errors are approximately normally distributed with constant variance; a heavily right-skewed target variable allows a small number of extreme large values to dominate the squared error, biasing the coefficient estimates.
- **Usage of this skill**: When the target skewness is greater than 1.0 and non-negative, it will automatically perform `log1p` transformation to compress the long tail and stabilize the variance; for missing value autofill, the median (more robust for skewed data) or mean (for symmetric distribution) will be selected according to the skewness.

5.2 Pearson Correlation Coefficient (Pearson correlation, r)

- **Meaning** : Measure the strength of **linear** correlation between two variables, take the value [-1,1], the closer | r | is to 1, the stronger the linear relationship.
- **Formula**: $r_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}$

- **Usage of this skill**: Step 4: Plot the correlation heatmap and the feature-target bar chart; Step 5: Eliminate features that have low correlation with the target (|r| < 0.03), and when two features are highly correlated(|r| > 0.9), retain the one that is more correlated with the target.
- **Note**: r only captures linear relationships; a strong non-linear correlation may also result in r ≈ 0.

5.3 Multicollinearity and VIF (Variance Inflation Factor)

- **Multicollinearity** refers to the phenomenon where multiple features are highly correlated with each other, leading to unstable regression coefficients (such as erratic sign flipping and extremely wide confidence intervals) and impairing interpretability.
- **VIF Formula**: Perform regression on the i -th feature using all other features to obtain the coefficient of determination R_i^2 , then $\mathrm{VIF}_i = \frac{1}{1 - R_i^2}$. R_i^2 The larger the value (the more linearly representable it is by other features), the larger the VIF, and the stronger the redundancy.
- **Intuitive Example**: If 90% of the variance of a given feature can be explained by other features ($R_i^2 = 0.9$), then $\mathrm{VIF} = 1/(1 - 0.9) = 10$, which means that the variance of its coefficient is magnified by approximately 10 times.
- **Empirical threshold**: VIF > 5 calls for attention, and VIF > 10 indicates severe collinearity. This skill adopts `VIF_THRESH = 10.0` . In Step 5, the variable with the highest VIF exceeding the threshold is iteratively removed until all variables meet the standard.

5.4 Cross-Validation (CV)

- **Purpose**: To evaluate only on the training set whether the model remains stable when switching to a different batch of data, so as to avoid overfitting caused by repeatedly tuning parameters using the test set.
- **K-fold CV (KFold)**: Divide the training set equally into K subsets, iteratively use K-1 subsets for training and the remaining 1 subset for validation, repeat this process K times and take the average of the resulting errors $\mathrm{CV} = \frac{1}{K} \sum_{k=1}^{K} \mathrm{Metric}_k$. For regular data in this skill, 4-fold shuffled KFold is adopted.
- **Time Series CV (TimeSeriesSplit)**: **random shuffling is not allowed**, otherwise data leakage will occur as future data is used to predict past data. The standard approach is to follow the time sequence: "use the earlier segment for training and the immediately subsequent segment for validation", with the window gradually expanding. For the time-series data in this skill, 4-fold TimeSeriesSplit is adopted.

- **Usage of this skill**: In Step 6, perform backward elimination based on the `--optimize-metric` on CV: try removing one feature at a time; if the CV metric improves, keep the feature removed, and repeat until no further improvement can be achieved.

5.5 Evaluation Metrics: Definition, Formula and Business Selection

Notation Convention: y_i ground truth, $\hat{y}_i$ predicted value, $\bar{y}$ mean of ground truth values, n number of samples, p number of features.

WMAPE (Weighted Mean Absolute Percentage Error, the default for this skill)

- **Formula**: $\mathrm{WMAPE} = \frac{\sum_{i=1}^{n}|y_i - \hat{y}_i|}{\sum_{i=1}^{n}|y_i|}$
- **Features**: With the total true value as the denominator, it is naturally weighted by volume; compared with MAPE, it is resistant to zero values and will not be amplified to infinity by a small denominator; the result is a dimensionless percentage, which facilitates cross-business comparison.

MAE (Mean Absolute Error)

- **Formula**: $\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$
- **Features**: Same dimension as the target value, most intuitive (e. g. "how many yuan on average"); equal weight assigned to each error, **insensitive to outliers**.

RMSE (Root Mean Square Error)

- **Formula**: $\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$
- **Features**: squaring first and then taking the square root, **amplifying the penalty for large errors**, most sensitive to outlier predictions; with the same dimension as the target.

R^2(coefficient of determination)

- **Formula**: $R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y}_i)^2}$, which refers to the proportion of variance explained by the model in the total variance, is $\leqslant 1$, and the closer it is to 1, the better.
- **Features**: Measures goodness of fit;**tends to be artificially inflated as the number of features increases** (it will not decrease even if irrelevant features are added).

Adjusted R-squared (Adjusted R^2)

- **Formula**: $\bar{R}^2 = 1 - (1 - R^2) \cdot \frac{n-1}{n-p-1}$
- **Features**: It imposes a penalty on the number of features p , and the more features there are, the heavier the penalty; it is more suitable than R^2 for **comparing models with different numbers of features**.

Which metric to optimize for which business scenario

Optimization Objective	Suitable Business Scenarios (Examples)
WMAPE	Amount / sales volume / GMV, etc. **with large magnitude differences, where overall deviation needs to be evaluated based on scale**: For example, when forecasting daily revenue of each store, the error of large stores is naturally more critical, and reporting in percentage terms is the most intuitive. It is also robust when there are many zero or small values.
MAE	When you want the error **to correspond one-to-one with business units and avoid being dominated by extreme values**: for relatively uniform targets such as forecasting customer service ticket volume and daily active users, stating "an average difference of N units" is directly communicable; this approach is more tolerant of occasional outliers in the data
RMSE	**When the cost of large errors is particularly high and outlier predictions must be suppressed as a priority**: for example, in inventory, production capacity or power load forecasting, a single severe underprediction or overprediction can lead to stockouts or resource waste, so heavier penalties need to be imposed on large deviations
R^2 / Adjusted R^2	when prioritizing **the overall explanatory power of the model and feature contributions** over absolute errors, such as in factor attribution or comparison of multiple feature schemes: when comparing models with different numbers of features, adjusted R^2 should be preferred to avoid being misled by redundant feature stacking

📌 This skill evaluation is based primarily on **the test set**, supplemented by the training set; although the severely right-skewed target is modeled on the log explicitness, all metrics are calculated on the original explicitness restored via **expm1** to ensure the errors carry business implications. The selected `--optimize-metric` will simultaneously guide the feature selection direction and be highlighted in the report.

5.6 Regression Coefficients, p-values and Confidence Intervals

- **Coefficient**: It indicates how much the target variable changes on average for each one-unit change in the feature; when standardized, it reflects the relative importance of the feature.
- **p-value**: Under the null hypothesis that "this feature has no effect on the target", it refers to the probability of observing the current (or more extreme) coefficient. **p < 0.05** is generally regarded as statistically significant (the effect is most likely not accidental), and a smaller p-value indicates higher reliability.
- **Confidence Interval (CI)**: a plausible range of values for a coefficient (e. g., 95% CI $\hat{\beta} \pm 1.96\,\mathrm{SE}(\hat{\beta})$); the narrower the interval, the more precise the estimate, **and if the interval crosses zero, the direction of the feature's effect is uncertain**.
- **Usage of this skill**: The report coefficient table provides coefficients, p-values and confidence intervals simultaneously, making it easy to determine which driving factors are both important and reliable.

5.7 How to Interpret Residual Diagnostic Plots

Residual $e_i = y_i - \hat{y}_i$ (actual value - predicted value). The residuals of an ideal linear model should **be randomly distributed around zero, free of structure, with constant variance and approximate normality**. This skill Step 7 outputs a residual diagnostic plot, and the method for interpreting the plot is as follows:

What are you looking at?	Health Performance (Model OK)	Abnormal Signal (Requires Handling)
Residuals vs. Predicted Values Scatter Plot	The point cloud is uniformly and randomly distributed above and below the 0 horizontal line, with no trend and no specific shape.	Shaped **like a funnel / horn** (more spread out on the right) = **heteroscedasticity**; curved / U-shaped = **nonlinearity** not captured
Whether the residuals are centered at zero	The overall mean is approximately 0, with roughly symmetrical positive and negative values	Systematically high or low = the model **is biased** (generally overestimating or underestimating)
Residual Distribution / Histogram	Approximately bell-shaped, unimodal and symmetric	Significant skewness or long tail = the target variable may require log transformation or contain outliers
Q-Q plot (if applicable)	The points roughly fall on the 45° reference line	Both ends tilting upward / deviation from the straight line = non-normal residuals with heavy tails

auto-feature-regression 详细使用文档

💡 **总结**：面向**回归预测任务**的自动化特征工程 + 线性回归建模 skill。给定业务背景和分析目标，输入一份原始数据集，自动产出预处理 / 特征工程 / 建模 / 评估的**中英双语报告** + 可视化图表 + 模型。

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34.81KB

一、使用场景

典型场景	说明
业务指标预测	预测日营收、订单量、GMV、成本、转化率等连续数值目标，并识别关键驱动因子
时间序列回归	含日期列的经营数据，需严格按时间划分训练 / 测试集、防止未来信息泄漏
快速可解释基线	需要一个系数可读、带 p 值 / 置信区间的线性模型作为 baseline，而非黑盒模型
特征工程探索	想快速试验交互项 / 多项式 / 时序衍生特征，并自动过滤掉无意义、共线、无增益的特征
分析报告产出	需要一份含描述统计、相关性热力图、冗余特征说明、模型公式与指标的报告

💡 **局限性：**

只能拟合线性回归模型，最后入模的特征都要求是数值类特征。

不适用于分类任务（目标为类别）、纯时序预测的深度模型、非结构化数据（图像 / 文本 / 音频）。

二、输入 / 工作流 / 输出

2.1 用户需要写的输入

必填参数

项	参数	说明
数据集文件	`--data`	CSV / Excel，含 1 个数值型目标列 + 若干特征列
目标字段名	`--target`	要预测的数值列名

关键可选参数

参数	何时用	示例
`--time-col`	当数据集是时间序列数据时，必传，否则退化为随机划分导致评估失真	`--time-col date`
`--exclude`	排除 ID / 名称等非特征列，或易泄漏的 target 衍生列	`--exclude id,shop_name`
`--context`	一句话业务背景，写入报告并让图解更贴切	`--context "广告季度收入数据"`
`--goal`	一句话建模目标，让图解围绕目标展开	`--goal "预测季度广告收入并找驱动因子"`
`--optimize-metric`	指定优化指标，驱动特征选择方向	WMAPE(默认)/MAE/RMSE/R2/Adj_R2
`--feature-mode`	特征构造模式	auto(默认)/custom/both
`--custom-features`	自定义业务特征公式	`"revenue_intensity = revenue_L30 / 30"`
`--interaction-pairs`	你清楚哪些特征相乘有业务含义时精确指定	`"price*volume; ad_cost*ctr"`
`--impute` / `--impute-overrides`	缺失值填充策略（全局 / 按列）	`--impute-overrides "recharge:zero"`
`--log-target`	严重右偏目标的 log1p 变换开关	auto(默认)/on/off
`--outdir`	输出目录	`--outdir ./output`

📝 **首次使用需安装依赖：**

```
pip install --user --break-system-packages pandas numpy scikit-learn matplotlib seaborn statsmodels tabulate
```

调用命令示例 —— 时间序列（必须传 `--time-col`）

时间序列建模

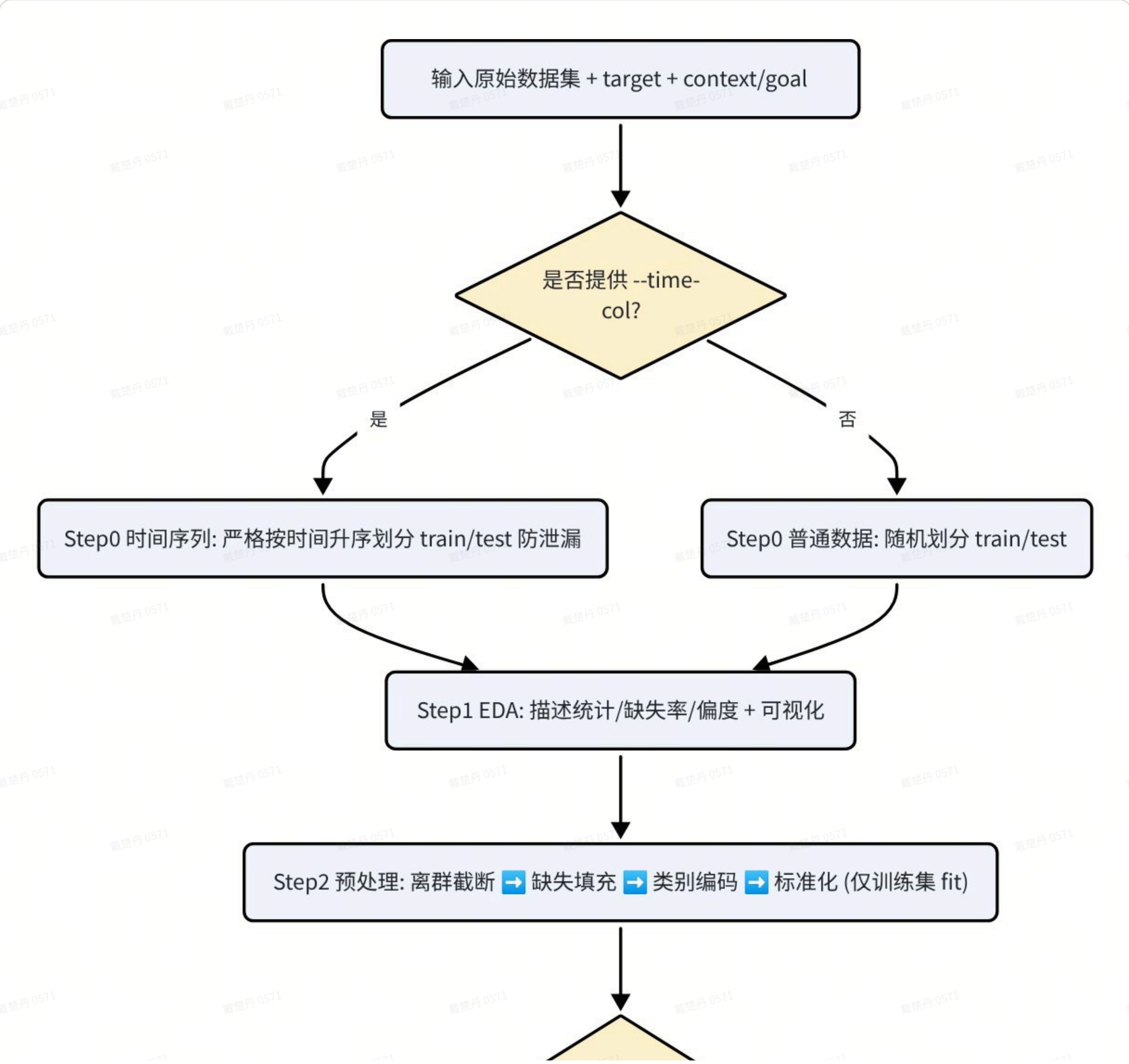
```
python3 scripts/auto_feature_regression.py \
    --data ~/files/sales.csv --target quarter_revenue \
    --time-col period_start_date --exclude account_id,account_name \
    --context "广告季度收入数据" --goal "预测季度广告收入并找驱动因子" \
    --optimize-metric WMAPE --outdir ./output
```

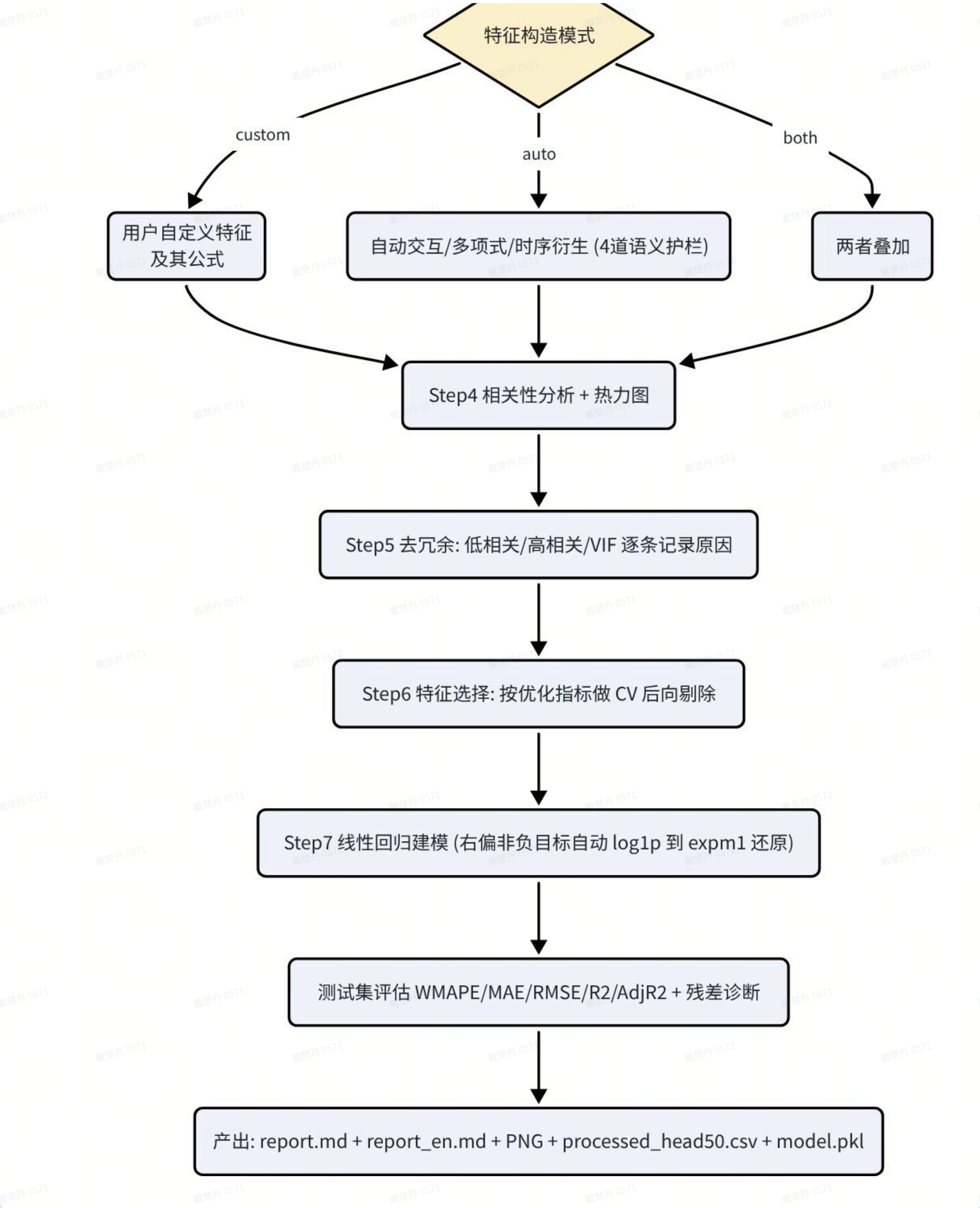
调用命令示例 —— 普通截面数据（不传 `--time-col` → 随机划分）

普通数据建模

```
python3 scripts/auto_feature_regression.py --data data.csv --target y --outdir ./output
```

2.2 工作流（Workflow）




特征构造模式
custom
auto
both
用户自定义特征及其公式
自动交互/多项式/时序衍生 (4道语义护栏)
两者叠加
Step4 相关性分析 + 热力图
Step5 去冗余: 低相关/高相关/VIF 逐条记录原因
Step6 特征选择: 按优化指标做 CV 后向剔除
Step7 线性回归建模 (右偏非负目标自动 log1p 到 expm1 还原)
测试集评估 WMAPE/MAE/RMSE/R2/AdjR2 + 残差诊断
产出: report.md + report_en.md + PNG + processed_head50.csv + model.pkl

7 步详解

Step	内容

Step 0 数据划分	提供 `--time-col` → 严格按时间升序切分（不打乱），防止未来信息泄漏；否则随机划分
Step 1 EDA	均值 / 方差 / 极值 / 偏度 / 缺失率 + 目标分布图或时序图
Step 2 预处理	离群分位截断（1% / 99%）→ **不因高缺失率删列**，按每列分布建议填充策略（中位数 / 均值 / 众数 / 填 0）或者直接强制全都将缺失值填充为0→ One-Hot → 标准化。所有 fit 参数只取自训练集
Step 3 特征构造	`auto` 自动造交互 / 多项式 / 时序衍生（交互项经四道护栏）； `custom` 用户给的业务公式； `both` 叠加。所有构造特征均登记，被跳过的组合逐条记录原因
Step 4 相关性分析	相关性热力图 + 特征-目标相关性条形图
Step 5 去冗余	与 target 低相关 / 特征间高相关 / VIF 多重共线性，逐条记录删除原因
Step 6 特征选择	以 `--optimize-metric` 为准，在训练集上做交叉验证后向剔除（时序 TimeSeriesSplit，普通 KFold，均 4 折）
Step 7 建模评估	线性回归 + 测试集指标 + 残差诊断；严重右偏非负目标自动 log1p、预测后 expm1 还原，指标在原始尺度计算

📌 **防数据泄漏是第一原则**：所有预处理参数（填充值、截断分位、编码、标准化）只在训练集上 fit，再应用到测试集；时间序列严格按时间切分，绝不打乱。

2.3 最后得到的输出（位于 --outdir）

文件	内容
`report.md`	分析背景目标、预处理汇总、描述统计、相关性、去冗余原因、构造特征清单、最终特征、模型公式 + 系数表（含 p 值 / 置信区间）、评价指标、数据划分说明；**每张图配数据驱动的「图解」**
`report_en.md`	上述报告的英文版，结构完全一致（叙述与表头翻译，图表 / 公式 / 数值共用），两版共享同一批图
`01_missing_rate.png` 等 PNG	缺失率、目标分布 / 时序、相关性热力图、特征-目标相关性条形图、残差诊断图；标题 / 坐标含义 / 单位均标注，大数值自动 K/M/B 格式化，无中文字体自动降级英文
`processed_head50.csv`	预处理后数据前 50 行，便于核对

`model.pkl`	拟合好的模型 + scaler + 特征列表 + 预处理参数，供后续复用 / 预测

2.4 示例

三、迭代过程记录

0723迭代

3.1 图表解释

结合 `--context` / `--goal` 与该图真实统计量（均值 / 偏度 / 最强相关 / 异方差迹象），说明这张图对分析目标意味着什么。

3.2 对目标做 log 变换（--log-target）

> ❓ **判定逻辑（auto 模式）**
>
> - `--log-target on` → 强制 log1p
> - `--log-target off` → 强制不变换
> - `--log-target auto`（默认）→ 当**目标偏度 > 1.0 且 目标最小值 ⩾ 0** 时，做 log1p 变换、预测后 expm1 还原，所有指标在原始尺度上计算；否则不变换。

为什么这样设计？ 收入、GMV、成本等严重右偏且非负的目标，取对数后分布更接近正态、更契合线性回归的等方差假设，能显著改善拟合与残差。含负值时自动关闭（log1p 无定义），保证鲁棒性。

3.3 交互项特征构造

原始版本盲目两两相乘，会产出大量无业务含义、加剧共线性的特征。现在 `auto` 模式对每个候选交互项 `a×b` 依次过闸，被跳过的组合逐条记录原因到报告：

		规则与阈值
①	同族过滤	`INTER_SKIP_SAME_FAMILY=True`：按列名词根识别「同一指标的不同期 / 分片」（如 revenue_q1 与 revenue_q2 词根都是 revenue），默认不相乘——同类相乘通常无业务含义。`--allow-same-family` 可放开
②	高共线过滤	`INTER_SKIP_HIGH_CORR=0.8`：特征相关系数 > 0.8 时不相乘，相乘只会进一步加剧共线性

③	增益校验	`INTER_MIN_GAIN=0.02`：交互项与 target 的 |corr| 须比两母特征各自 |corr| 至少高出 0.02，否则视为无信息增量丢弃
④	数量上限	`INTER_MAX_KEEP=8`：按增益排序只保留 Top-8，防维度爆炸

💡 若用户清楚哪些相乘才有业务含义，用 `--interaction-pairs "price*volume;ad_cost*ctr"` 精确指定 → **跳过全部自动组合与护栏**，只造用户列出的对。

3.4 缺失值处理

- **不再因高缺失率删列**：缺失率超过 `HIGH_MISS_HINT(=0.5)` 仅作提示，保留由人工判断信息价值。
- `--impute auto` 基于每列分布特征（偏度、是否非负、0 占比、基数）自动建议策略（中位数 / 均值 / 众数 / 填 0），可用 `--impute` 全局指定或 `--impute-overrides` 按列覆盖。

3.5 去冗余与特征选择规则

- **低相关剔除**：与 target 的 |r| < `CORR_TARGET_MIN(=0.03)` 的候选剔除。
- **高相关剔除**：两特征 |r| > `CORR_HIGH(=0.9)`，保留与 target 更相关者。
- **VIF 剔除**：迭代删除 VIF > `VIF_THRESH(=10.0)` 的特征。
- **CV 后向剔除**：以 `--optimize-metric` 为准，逐一尝试移除某特征，若能改善交叉验证指标则移除，直至无法改善。

四、未来优化方向

⭐ 为适应更多场景，后续可从以下方向增强本 skill。

1. **模型族扩展**：加入 Ridge / Lasso / ElasticNet（自动正则化、内置特征选择），以及可解释树模型（如带 SHAP 的 LightGBM）作为可选 baseline，对比线性与非线性拟合能力。
2. **分类 / 计数任务支持**：扩展到逻辑回归、泊松 / 负二项回归，覆盖二分类、比率、计数型目标，让 skill 不局限于连续回归。
3. **更丰富的特征构造**：加入分箱 / 目标编码、周期性时间特征（星期 / 月 / 节假日 one-hot）、多阶交互（三元），以及基于业务词典的自动语义分组。
4. **超参与阈值自动调优**：把 `Config` 中的截断分位、VIF / 相关性阈值、交互增益门槛做成可网格 / 贝叶斯搜索的对象，按数据自适应而非固定值。

5. **数据质量与漂移检测**：新增训练 / 测试分布漂移检测（PSI / KS）、异常样本标记、目标泄漏自动检测（与 target 相关性异常高的特征自动预警）。
6. **交叉验证与稳健性增强**：支持分组 CV（按店铺 / 用户）、嵌套 CV、时间滚动预测评估；输出预测区间而非仅点估计。
7. **交互式与可复用产物**：输出可交互 HTML 报告、一键预测脚本（加载 `model.pkl` 对新数据打分）、以及 REST 化的预测接口模板。

五、涉及的统计学知识

> 💡 本节是skill 全流程涉及的统计学知识：概念含义、**计算公式**、**业务选型建议**与**看图方法**，帮助非统计背景的读者读懂报告里每个指标与判定。

线性回归的假设LINE

> 🥛 L是Linear(线性)，指的是因变量与自变量之间的关系应该满足线性关系；I是Independent(独立性)，指的是自变量之间相互独立，残差之间相互独立，残差和自变量之间相互独立；N是Normality(正态性)，指的是残差的分布应该服从正态分布；E是Equal Variance(方差齐性)，即残差的方差是常数。

5.1 偏度（Skewness）与 log 变换

- **含义**：衡量分布相对正态的**不对称程度**。偏度 > 0 右偏（长尾在右，少数大值，如收入 / GMV），< 0 左偏，≈ 0 近似对称。
- **公式**：$\text{Skew} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{\sigma}\right)^3$，即标准化后三阶矩。
- **为何关注**：线性回归假设误差近似正态、方差恒定；严重右偏目标让少数极端大值主导平方误差，拉偏系数估计。
- **本 skill 用法**：目标偏度 > 1.0 且非负时自动 `log1p` 变换，压缩长尾、稳定方差；缺失填充也据偏度选中位数（偏态更稳健）或均值（对称）。

5.2 皮尔逊相关系数（Pearson correlation, r）

- **含义**：衡量两变量**线性**相关强度，取值 [-1, 1]，|r| 越接近 1 线性关系越强。
- **公式**：$r_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}$
- **本 skill 用法**：Step 4 画相关性热力图与特征-目标条形图；Step 5 剔除与 target 低相关（|r| < 0.03）的特征，两特征高相关（|r| > 0.9）时保留与 target 更相关者。

- **注意**：r 只捕捉线性关系，非线性强相关也可能 r ≈ 0。

5.3 多重共线性与 VIF（方差膨胀因子）

- **多重共线性**：多个特征彼此高度相关，导致回归系数不稳定（符号乱跳、置信区间极宽），削弱可解释性。
- **VIF 公式**：把第 i 个特征用其余所有特征做回归，得到判定系数 R_i^2，则 $\mathrm{VIF}_i = \dfrac{1}{1-R_i^2}$。$R_i^2$ 越大（越能被其他特征线性表示）→ VIF 越大 → 冗余越强。
- **直觉举例**：若某特征 90% 的方差能被其他特征解释（$R_i^2 = 0.9$），则 $\mathrm{VIF} = 1/(1-0.9) = 10$，意味着其系数方差被放大约 10 倍。
- **经验阈值**：VIF > 5 需警惕，> 10 判定严重共线。本 skill 取 `VIF_THRESH = 10.0`，Step 5 迭代删除 VIF 最高且超阈值者，直至全部达标。

5.4 交叉验证（Cross-Validation, CV）

- **目的**：只在训练集上评估「换一批数据模型是否仍稳定」，避免用测试集反复调参而过拟合。
- **K 折 CV（KFold）**：训练集均分 K 份，轮流用 K-1 份训练、剩 1 份验证，重复 K 次取平均误差 $\mathrm{CV} = \dfrac{1}{K}\sum_{k=1}^{K}\mathrm{Metric}_k$。本 skill 普通数据用 4 折 KFold（打乱）。
- **时间序列 CV（TimeSeriesSplit）**：**不能随机打乱**，否则用未来预测过去造成泄漏。做法是按时间递增「用前段训练、验证紧随其后的一段」，窗口逐步扩大。本 skill 时序数据用 4 折 TimeSeriesSplit。
- **本 skill 用法**：Step 6 以 CV 上的 `--optimize-metric` 为准做后向剔除——逐一尝试移除某特征，若 CV 指标改善则移除，直至无法改善。

5.5 评价指标：含义、公式与业务选型

符号约定：y_i 真实值，$\hat{y}_i$ 预测值，$\bar{y}$ 真实值均值，n 样本数，p 特征数。

WMAPE（加权平均绝对百分比误差，本 skill 默认）

- **公式**：$\mathrm{WMAPE} = \dfrac{\sum_{i=1}^{n}|y_i - \hat{y}_i|}{\sum_{i=1}^{n}|y_i|}$
- **特点**：以真实值总量为分母，天然按体量加权；相比 MAPE 抗零值、不会被小分母放大成无穷大；结果是无量纲百分比、便于跨业务比较。

MAE（平均绝对误差）

- **公式**：$\mathrm{MAE} = \dfrac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$

- **特点**：与目标同量纲、最直观（"平均差多少元"）；对每个误差等权，**对异常值不敏感**。

RMSE（均方根误差）

- **公式**：$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$
- **特点**：先平方再开方，**放大大误差的惩罚**，对离群预测最敏感；与目标同量纲。

R^2（判定系数）

- **公式**：$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y}_i)^2}$，即模型解释的方差占总方差比例，⩽ 1，越接近 1 越好。
- **特点**：衡量拟合优度；**会随特征增多而虚高**（哪怕加无关特征也不降）。

调整后 R^2（Adjusted R^2）

- **公式**：$\bar{R}^2 = 1 - (1 - R^2) \cdot \frac{n-1}{n-p-1}$
- **特点**：对特征数量 p 做惩罚，特征越多惩罚越重；比 R^2 更适合**比较不同特征数量**的模型。

什么业务场景优化哪个指标

优化目标	适合的业务场景（举例）
WMAPE	金额 / 销量 / GMV 等**量级差异大、要按体量看整体偏差**的业务：如预测各门店日营收，大店误差自然更重要，用百分比汇报最直观。含较多零值或小值时也稳健。
MAE	希望误差**与业务单位一一对应、且不想被极端值主导**时：如预测客服工单量、日活等相对均匀的目标，"平均差 N 个"直接可沟通；对数据中偶发异常点更宽容
RMSE	**大误差代价特别高、必须重点压制离群预测**时：如库存 / 产能 / 电力负荷预测，一次严重低估或高估会造成断货或浪费，需要对大偏差施加更重惩罚
R^2 / 调整后 R^2	更关心**模型整体解释力与特征贡献**而非绝对误差时：如做因子归因、对比多套特征方案。特征数不同的模型对比优先用调整后 R^2，避免被特征堆砌误导

> 📌 本 skill 评估以**测试集**为主、附训练集;严重右偏目标虽在 log 尺度建模，但所有指标均在 **expm1 还原后的原始尺度**上计算，保证误差有业务含义。选定的 `--optimize-metric` 会同时驱动特征选择方向并在报告中高亮。

5.6 回归系数、p 值与置信区间

- **系数（coefficient）**：特征每变动一个单位、目标平均变动多少;已标准化时反映相对重要性。

- **p 值**：在「该特征对目标无影响」的原假设下，观测到当前（或更极端）系数的概率。**$p < 0.05$** 通常视为统计显著（效应大概率非偶然），p 越小越可信。
- **置信区间（CI）**：系数的合理取值区间（如 95% CI $\hat{\beta} \pm 1.96\,\mathrm{SE}(\hat{\beta})$）;区间越窄估计越精确，**若区间跨越 0 则该特征效应方向不确定**。
- **本 skill 用法**：报告系数表同时给出系数 / p 值 / 置信区间，便于判断哪些驱动因子既重要又可信。

5.7 残差诊断图怎么看

残差 $e_i = y_i - \hat{y}_i$（真实 - 预测）。理想线性模型的残差应**随机分布在 0 附近、无结构、方差恒定、近似正态**。本 skill Step 7 输出残差诊断图，读图方法：

看什么	健康表现（模型 OK）	异常信号（需处理）
残差 vs 预测值 散点	点云均匀随机地散布在 0 水平线上下，无趋势、无形状	呈**漏斗形 / 喇叭形**（右侧越散）= **异方差**;呈曲线 / U 形 = 存在**非线性**未被捕捉
残差是否居中于 0	整体均值 ≈ 0，正负大致对称	系统性偏上或偏下 = 模型**有偏**（普遍高估或低估）
残差分布 / 直方图	近似钟形正态、单峰对称	明显偏斜或长尾 = 目标可能需 log 变换或存在离群
Q-Q 图（若有）	点大致落在 45° 参考线上	两端翘起 / 偏离直线 = 残差非正态、有厚尾